Quantitative Empirical Methods Exam

Yale Department of Political Science, August 2024

You have 24 hours to complete the exam. The exam consists of three parts.

Back up your assertions with mathematics where appropriate and show your work. Good answers will provide a direct answer that illustrates an understanding of the question, and calculations or statistical arguments to validate the answer. Where applicable, exceptional answers will include all of these *as well as* proofs that are technically complete, including formally articulating sufficient assumptions and regularity conditions. Questions will not be weighted equally. A holistic score will be assigned to the exam. Therefore, it is important to demonstrate your understanding of the material to the best of your ability.

Part 1 (Theoretical section) consists of short questions that can be answered with pen and paper. You are allowed to consult textbooks and other reference material, but the questions are written so that well-prepared students should be able to answer them without such references.

Part 2 (Essay section) contains a recent, well-regarded empirical article. We will ask you to offer an evaluation of its methodological approach and presentation of results. In particular, we will advise you to pay particular attention to the identification conditions (either explicit or implicit), the associated estimation strategy, and possible threats to inference. Your response may be anywhere from 500 to 1,000 words.

Part 3 (Computer assisted section) will involve using statistical software to answer one longer exercise with several associated questions. A complete answer to Part 3 will include code and output, as well as your written answers.

For the whole exam, you are permitted access to any and all written materials, as well as unrestricted use of your own computer with access to the internet. The only restriction is that you may **not** interact with any person, online or otherwise.

Please turn in your answers as an email to colleen.amaro@yale.edu.

1 Theoretical section

1. Provide brief definitions of the following terms:
   (a) bias
   (b) bias-variance tradeoff
   (c) confidence interval
   (d) identification
   (e) p-value
   (f) statistical power

2. Consider an experiment with n total units, in which treatment has been assigned using complete randomization. Researchers often estimate the average treatment effect within an experiment using a difference-in-means estimator.

$$\hat{\tau}_{DiM} = \frac{\sum_{i=1}^n T_i Y_i}{\sum_{i=1}^n T_i} - \frac{\sum_{i=1}^n (1 - T_i) Y_i}{\sum_{i=1}^n (1 - T_i)},$$

   where T_i is an indicator for whether or not an individual is assigned treatment.

   (a) Show that $\hat{\tau}_{DiM}$ is an unbiased estimator for the ATE.
   (b) An alternative approach is to estimate the ATE by regressing the outcomes with the treatment assignment indicator (represented as $\hat{\tau}_{ols}$). Show that the estimated ATE from running the regression $Y \sim T$ will be equivalent to estimating the difference-in-means (i.e., $\hat{\tau}_{ols} \equiv \hat{\tau}_{DiM}$).
   (c) In part (a), you showed that the difference-in-means estimator is an unbiased estimator for the ATE within an experiment. Now consider the setting in which you are interested in not only the ATE across the experimental sample, but also, the ATE across a target population. Let $S_i = 1$ for units in the experimental sample, and let $S_i = 0$ be the units in the target population. The estimand of interest is the PATE (i.e., $\mathbb{E}(Y_i(1) - Y_i(0) \mid S_i = 0)$). What is the necessary assumption for the difference-in-means estimator to be unbiased for the PATE?

3. Consider again an experimental setting. Researchers have collected a set of pre-treatment covariates $\mathbf{X}$.

   (a) What is one way in which researchers could incorporate these covariates into their experimental design to improve the efficiency of their ATE estimation *at the design stage*?

(b) Now assume researchers have access to a set of pre-treatment covariates, but cannot incorporate them at the design stage. How might researchers incorporate $\mathbf{X}$ into their estimation in a post-hoc manner to improve the efficiency of their ATE estimates?

(c) Explain when researchers should expect there to be the largest improvements in efficiency.

4. You conduct a field experiment in which you want to evaluate the impact of door-to-door canvassing on voter turnout. Individuals are randomly assigned to either a treatment group that is attempted with a door knock or to an uncontacted control group. About 20% of people in the treatment group did not open the door.

   (a) Can you identify the average treatment effect?

   (b) In settings with non-compliance, the estimand of interest is often the intent to treat effect (ITT) or the complier average causal effect (CACE). A common approach that is used in practice to estimate the CACE is the use of instrumental variables. What is the instrument in this setting? Being as specific as possible, how would one use instrumental variables to estimate the CACE in this setting? Write down a model.

   (c) What assumption(s) needs to hold in order for the IV estimator to be an unbiased estimator for the CACE? Given this setting, what might make these assumption(s) plausible or not plausible? How would you investigate these assumption(s)?

   (d) How might we expect the CACE to differ from the ITT?

5. A colleague conducted a survey experiment on Lucid. Respondents were randomly assigned to an untreated control group or a treatment group that read an article where an individual shares an emotional, personal experience about how climate change is harming their community. Next, all respondents were informed that Members of Congress were considering new legislation (a carbon tax) to address the impacts of climate change and offered them a chance to share their thoughts on this policy idea with their Member of Congress. Six trained coders who were blind to the treatment condition hand-coded these messages as either pro-carbon tax or anti-carbon tax. The trained coders also coded respondents who were unsure of their opinion, respondents who stated that they had no opinion or didn't want to share their opinion, responses that were unclear, and those that were gibberish. In all cases, the coders were in perfect agreement with one another. Your colleague is very excited about this behavioral outcome measure.

   To conduct their analysis, your colleague recodes this data so respondents who write in support of taking action on climate change get a 1 and those who write in opposition get a 0. Your colleague finds that 32% of respondents in the control group and 57% of respondents in the treatment group who chose to write either a pro- or anti- message did so in support of taking action on climate change ($p < 0.001$). Your colleague claims to you that this provides strong evidence that reading an emotional personal story about the harms of climate change can mobilize political action taking.

Your colleague also shares with you that they tested for covariate balance and differential attrition. You are satisfied by the tests they share with you.

As a methodologist, what advice do you give your colleague about their analysis before they submit this paper to a journal?

6. Suppose a random variable X has mean μ.

   (a) Provide an example of an estimator of μ that is biased for μ but is consistent for μ.

   (b) Provide an example of an estimator of μ that is unbiased for μ but is inconsistent for μ.

2 Essay section

Read the following article and offer a critical evaluation of its methodological approach and presentation of results. Note: "critical" does not imply that you must only criticize – it is recommended that you give credit to the authors if and when their arguments are convincing and/or novel with respect to standard practice.

Your response should be between 500 to **1000** words. We advise you to pay particular attention to the identification conditions (either explicit or implicit), the associated estimation strategy, possible threats to inference, statistical power, and generalizability. Justify each of your claims and, where applicable, suggest ways in which this line of research might be improved. We do not expect you to have special expertise in the topic area, but we do expect you to bring to bear your general analytical skills as a political scientist.

Article:

> Harding, Robin, and Arinze Nwokolo. "Terrorism, trust, and identity: Evidence from a natural experiment in Nigeria." *American Journal of Political Science* (2024). `https://doi.org/10.1111/ajps.12769`

3 Computer assisted section

In this problem you are asked to estimate confidence intervals for a parameter of interest using several different procedures.

Let $\tau = \frac{\mu_Y(1)-\mu_Y(0)}{\mu_Y(0)}$ be the parameter of interest. A randomized experiment was conducted, and $\bar{Y}_t$ (average outcome in the treated group) and $\bar{Y}_c$ (average outcome in the control group) were calculated. We will use the Lalonde NSW data. The outcome variable is *re78* and the treatment variable is *treat*. You can load the data using:

```
library(Matching)
data(lalonde)
```

In this question we will use several different procedures to construct a confidence interval/set for τ from the data. An intuitive estimator for τ is:

$$\hat{\tau} = \frac{\bar{Y}_t - \bar{Y}_c}{\bar{Y}_c} \tag{1}$$

1. Use bootstrap to construct a 5% confidence interval for τ. (Please set.seed(12345) to make your procedure replicable.)
2. Assume the following model for the treatment effect:

   $$Y(1) = Y(0) * \delta \tag{2}$$

   Construct a 5% confidence interval for τ using permutation inference. (Use 300 repetitions for constructing the permutation distribution using Monte-Carlo simulations.) What is your "best guess" for τ? Is it similar to your estimate of $\hat{\tau}$?
3. Assume the following model for the treatment effect:

   $$Y(1) = Y(0) + \delta \tag{3}$$

   Construct a 5% confidence interval for τ using permutation inference. (Use 300 repetitions for constructing the permutation distribution using Monte-Carlo simulations.) What is your "best guess" for τ? Is it similar to your estimate of $\hat{\tau}$?
4. Is there a way to test whether the additive treatment effect model fits the data? In other words, does the additive treatment effect model have testable implication(s) which can be tested in the data? If yes, explain and test whether the testable implication(s) are satisfied in the data.

5. Assume that $Y_i(0) \sim N(\mu_c, \sigma^2)$ and $Y_i(1) \sim N(\mu_t, \sigma^2)$, and therefore:

$$Y_i(1) - Y_i(0) \sim N(\mu_t - \mu_c, 2\sigma^2). \tag{4}$$

   Can we use the normality assumption to construct a confidence interval for τ without using asymptotic theory, i.e. without using limiting distributions? If yes, construct a confidence interval for τ at a 5% level. If it is not possible, prove formally why not.

6. Use the delta method to construct a confidence interval for τ using the asymptotic distribution of $\hat{\tau}$. You can can assume that $\frac{n_{treat}}{n_{control}} \xrightarrow{p} k$, i.e., the ratio between the treatment and control group converges to a constant, $0 < k < 1$.

7. Compare and discuss the results from all the different procedures. Which one will you choose as a researcher and why? (Limit your answer to no more than half a page.)